

Engineering Programming II: Possible Topics

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1 Intro

1.1 IDEs

- i. Anaconda3
- ii. Spyder IDE

2 Language and syntax

2.1 Objects

- i. Data types: int, float, bool, str
- ii. Variables and literals
- iii. type()
- iv. Casting

2.2 Operators

- i. Numeric
- ii. Boolean
- iii. String

2.3 Lists and dictionaries

- i. Slicing, appending
- ii. Multi-dimensional lists

2.4 Loops

- i. For
- ii. While

2.5 Conditionals

- i. If-else
- ii. if-elif-else

2.6 User-defined functions

- i. def, return
- ii. Positional/keyword arguments
 - i. join, split, strip, upper, lower,

2.7 More strings

2.8 Importing packages

- i. import, as
- ii. math, random, os (getcwd, listdir), time (sleep, time)

2.9 File I/O

2.10 Error handling

3 Scientific packages

3.1 NumPy

- i. math
- ii. arrays: sorting, argmax, where, argwhere, ones, zeros, eye, rand, randn, identity, cumsum, all, any, isinf, isnan,
- iii. histogram, bincount
- iv. stats: mean, mediand, std,
- v. random
- vi. linalg
- vii. fft

3.2 Matplotlib

- i. plot, hist, imshow, scatter
- ii. fig plt ax
- iii. subplots

3.3 SciPy

- i. odes
- ii. optimization

4 Computer arithmetic

4.1 Data types

4.2 Floating point representation

4.3 Integer division

4.4 Error

5 Random variables, RNG, and Statistics

5.1 Probability distributions

5.2 Law of Large Numbers (std of mean, integral of f over p)

5.3 Random number generation

5.4 numpy.random

- 5.5 Random walk
- 5.6 Drawing from a custom distribution (linear interpolation/binning)
- 6 Monte Carlo simulation**
 - 6.1 Calculating π
 - 6.2 Calculating integrals
 - 6.3 Functions of distributions
- 7 Linear Algebra (solving linear systems)**
 - 7.1 numpy.linalg
 - 7.2 Condition number
- 8 Root-finding (solving nonlinear systems)**
 - 8.1 Newton's method
- 9 Regression**
 - 9.1 Pearson correlation coefficient
 - 9.2 Linear Least squares
- 10 Optimization**
 - 10.1 Calculus review: $df/dx = 0$
 - 10.2 Gradient descent
 - 10.3 Simulated annealing
 - 10.4 Stochastic gradient descent
 - 10.5 Genetic algorithms
- 11 Fourier Analysis**
- 12 ODEs (IVPs)**
 - 12.1 Discretization
 - 12.2 Methods
 - i. Euler's method (1st order)
 - ii. Heun's method (2nd order)
 - iii. Runga-Kutta (4th order)
 - iv. SciPy ODE solvers
- 12.3 Examples
 - i. Exponential growth
 - ii. Projectile motion
 - iii. Lotka-Volterra predator-prey
 - iv. Simple pendulum
 - v. n -body problem
- 13 PDEs (time-dependent)**
 - 13.1 Discretization
 - 13.2 Methods
 - i. Forward difference
 - ii. Backward difference
 - iii. Crank-Nicholson
 - iv. Spectral
 - 13.3 Examples
 - i. Advection
 - ii. Wave equation
 - iii. Schrodinger equation
- 14 Agent-based modeling**
 - 14.1 Cellular automata
 - 14.2 Conway's game of life
 - 14.3 SIR pandemic modeling
- 15 Markov chains**
 - 15.1 Simple example
 - 15.2 Definition
 - 15.3 NLP n -grams
 - 15.4 Markov chain Monte Carlo methods